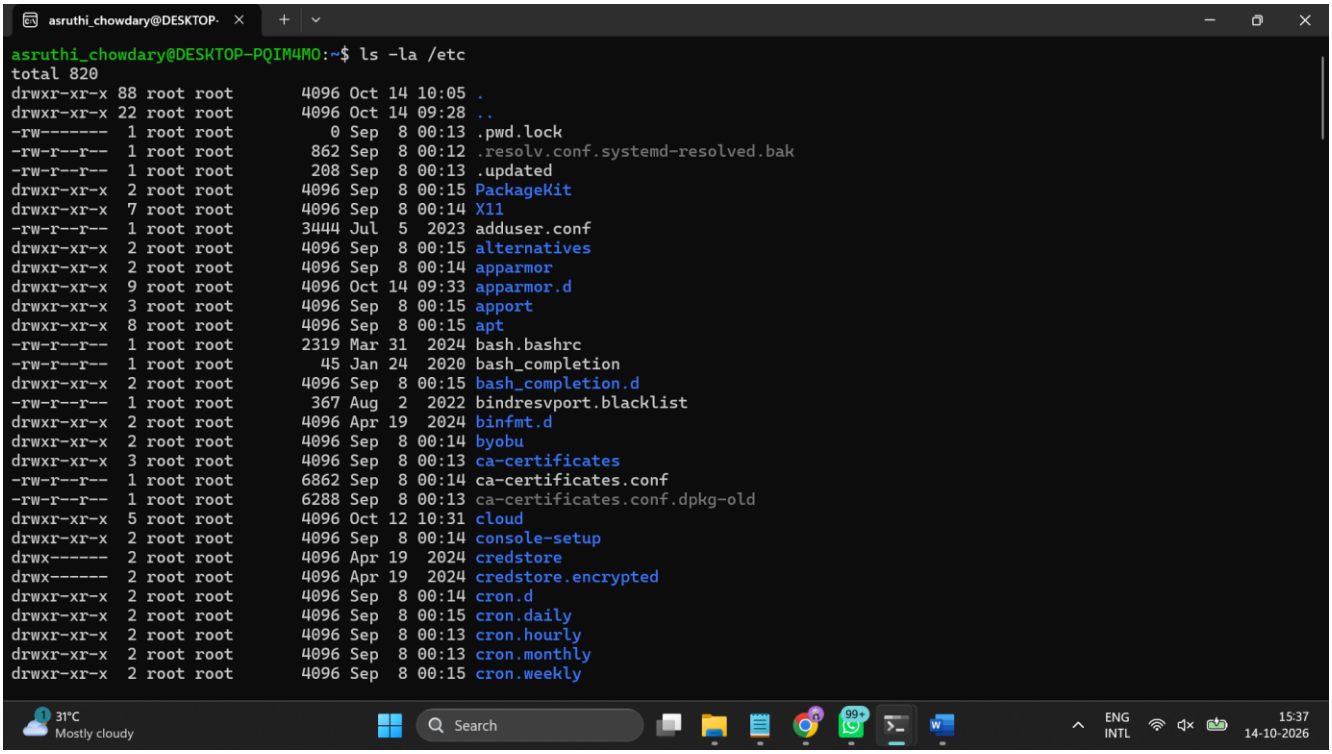
1. Linux File System Explorer

Goal: Explore common Linux directories and understand where configuration, logs, user files, temporary files, and programs are stored.

1.1 — Explore /etc

Command: `ls -la /etc`

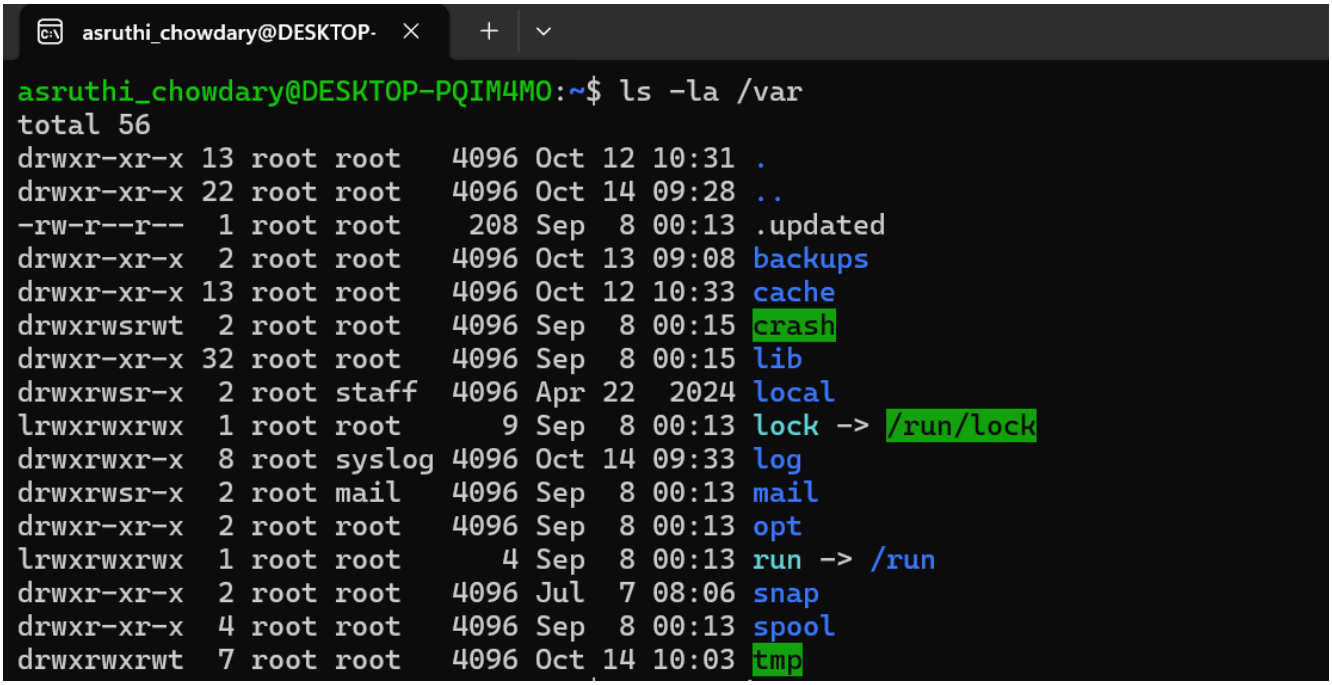
Explanation: Lists configuration files and directories under /etc.



1.2 — Explore /var

Command: `ls -la /var`

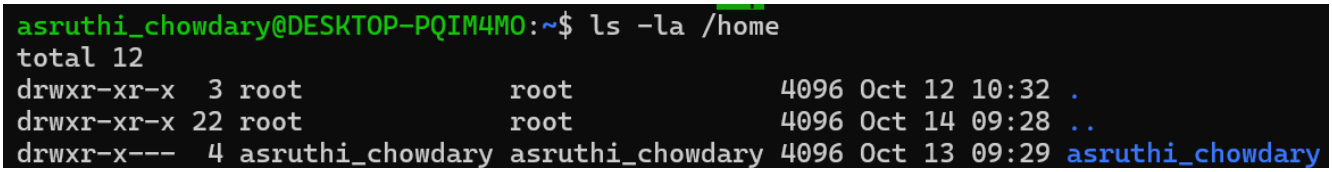
Explanation: Lists variable data directories; logs and other changing application data are commonly stored under /var.



1.3 — Explore /home

Command: `ls -la /home`

Explanation: Lists home directories for regular users.



1.4 — Explore /tmp

Command: `ls -la /tmp`

Explanation: Lists temporary files; these may be removed automatically by the system.



1.5 — Explore /usr

Command: `ls -la /usr`

Explanation: Lists software, libraries, documentation, and other user-space resources.

```
asruthi_chowdary@DESKTOP-PQIM4MO:~$ ls -la /usr
total 72
drwxr-xr-x  12 root root  4096 Sep  8 00:13 .
drwxr-xr-x  22 root root  4096 Oct 14 09:28 ..
drwxr-xr-x   2 root root 28672 Oct 14 09:33 bin
drwxr-xr-x   2 root root  4096 Apr 22  2024 games
drwxr-xr-x   4 root root  4096 Oct 14 09:33 include
drwxr-xr-x  56 root root  4096 Oct 12 10:31 lib
drwxr-xr-x   2 root root  4096 Oct 14 09:33 lib64
drwxr-xr-x   7 root root  4096 Oct 14 09:33 libexec
drwxr-xr-x  10 root root  4096 Sep  8 00:13 local
drwxr-xr-x   2 root root  4096 Oct 14 09:33 sbin
drwxr-xr-x 109 root root  4096 Sep  8 00:14 share
drwxr-xr-x   2 root root  4096 Apr 22  2024 src
```

2. Linux Commands Investigation

Goal: Practise navigation, listing files, creating directories, and understanding safe deletion.

2.1 — Print working directory

Command: `pwd`

Explanation: Prints the full path of the directory you are currently in.

2.2 — List files normally

Command: `ls`

Explanation: Lists visible, non-hidden entries in the current directory.

2.3 — List all details

Command: `ls -la`

Explanation: Shows hidden entries and a long listing with permissions, ownership, size, and modification time.

2.4 — Create a practice directory

Command: `mkdir -p ~/linux_lab/commands`

Explanation: Creates the directory path if it does not already exist.

2.5 — Create a test file

Command: `touch ~/linux_lab/commands/sample.txt`

Explanation: Creates an empty file if it does not exist, or updates its modification time.

2.6 — Remove a test file safely

Command: `rm ~/linux_lab/commands/sample.txt`

Explanation: Removes the specified file; check the path carefully because rm normally does not use a recycle bin.

```
asruthi_chowdary@DESKTOP-PQIM4MO:~$ pwd
/home/asruthi_chowdary
asruthi_chowdary@DESKTOP-PQIM4MO:~$ ls
asruthi_chowdary@DESKTOP-PQIM4MO:~$ ls -la
total 32
drwxr-x--- 4 asruthi_chowdary asruthi_chowdary 4096 Oct 13 09:29 .
drwxr-xr-x 3 root             root             4096 Oct 12 10:32 ..
-rw------- 1 asruthi_chowdary asruthi_chowdary 1258 Oct 13 09:46 .bash_history
-rw-r--r-- 1 asruthi_chowdary asruthi_chowdary  220 Oct 12 10:32 .bash_logout
-rw-r--r-- 1 asruthi_chowdary asruthi_chowdary 3771 Oct 12 10:32 .bashrc
drwx------ 2 asruthi_chowdary asruthi_chowdary 4096 Oct 12 10:33 .cache
-rw-rw-r-- 1 asruthi_chowdary asruthi_chowdary    0 Oct 14 09:28 .motd_shown
-rw-r--r-- 1 asruthi_chowdary asruthi_chowdary  807 Oct 12 10:32 .profile
drwx------ 2 asruthi_chowdary asruthi_chowdary 4096 Oct 12 10:36 .ssh
-rw-r--r-- 1 asruthi_chowdary asruthi_chowdary    0 Oct 13 09:29 .sudo_as_admin_successful
asruthi_chowdary@DESKTOP-PQIM4MO:~$ cd /temp
-bash: cd: /temp: No such file or directory
asruthi_chowdary@DESKTOP-PQIM4MO:~$ mkdir -p ~/linux_lab/commands
asruthi_chowdary@DESKTOP-PQIM4MO:~$ touch ~/linux_lab/commands/sample.txt
asruthi_chowdary@DESKTOP-PQIM4MO:~$ rm ~/linux_lab/commands/sample.txt
asruthi_chowdary@DESKTOP-PQIM4MO:~$
```

3. Linux Permissions

Goal: Inspect permission bits and practise changing permissions on a test file only.

3.1 — Inspect current permissions

Command: `ls -l ~/linux_lab/commands`

Explanation: Shows permissions, owner, group, size, and modification time for entries in the directory.

3.2 — Create a permissions test file

Command: `touch ~/linux_lab/commands/permission-demo.txt`

Explanation: Creates an empty file for permission experiments.

3.3 — Set permissions to 644

Command: `chmod 644 ~/linux_lab/commands/permission-demo.txt`

Explanation: Gives the owner read/write and group/others read-only permissions.

3.4 — Verify 644 permissions

Command: `ls -l ~/linux_lab/commands/permission-demo.txt`

Explanation: Displays the permission string; 644 normally appears as -rw-r--r-- for a regular file.

3.5 — Set permissions to 755

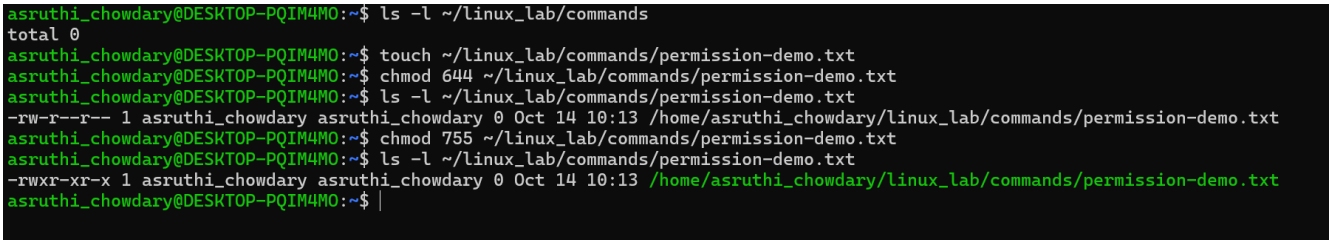
Command: `chmod 755 ~/linux_lab/commands/permission-demo.txt`

Explanation: Gives the owner read/write/execute and group/others read/execute; execute permission is usually unnecessary for an ordinary text file.

3.6 — Verify 755 permissions

Command: `ls -l ~/linux_lab/commands/permission-demo.txt`

Explanation: Displays the permission string; 755 normally appears as -rwxr-xr-x for a regular file.



4. Linux Processes

Goal: Inspect processes and observe basic resource usage. Output can differ between systems.

4.1 — Show processes attached to this terminal

Command: `ps`

Explanation: Shows a snapshot of processes associated with the current terminal/session.

4.2 — Show processes for all users

Command: `ps aux`

Explanation: Shows a detailed snapshot of processes across users, including CPU and memory usage.

4.3 — Monitor processes interactively

Command: `top`

Explanation: Displays a live, updating view of processes and CPU/memory activity; press q to quit.

4.4 — Search for a process by name

Command: `ps aux | grep -i ssh`

Explanation: Filters the process listing for lines containing “ssh”; the grep command itself may also appear.

4.5 — View memory summary

Command: `free -h`

Explanation: Shows RAM and swap usage in human-readable units; useful alongside process inspection.ps

```
asruthi_chowdary@DESKTOP-PQIM4MO:~$ ps
    PID TTY          TIME CMD
   3761 pts/2    00:00:00 bash
   3835 pts/2    00:00:00 ps
asruthi_chowdary@DESKTOP-PQIM4MO:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.1  0.3  22036 13188 ?        Ss   09:26   0:03 /usr/lib/systemd/systemd --system --deserialize=54
root           2  0.0  0.0   3180  2204 hvc0     Sl+  09:26   0:00 /init
root           6  0.0  0.0   3180  2044 hvc0     Sl+  09:26   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-f
root          48  0.0  0.4  66840 15984 ?        S<s  09:26   0:02 /usr/lib/systemd/systemd-journald
root         105  0.0  0.1  25148  6496 ?        Ss   09:26   0:01 /usr/lib/systemd/systemd-udevd
systemd+     116  0.0  0.3  21348 13180 ?        Ss   09:26   0:00 /usr/lib/systemd/systemd-resolved
systemd+     117  0.0  0.2  91040  7944 ?        Ssl  09:26   0:00 /usr/lib/systemd/systemd-timesyncd
root         169  0.0  0.0   4244  2684 ?        Ss   09:26   0:00 /usr/sbin/cron -f -P
message+     170  0.0  0.1   9780  5664 ?        Ss   09:26   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile
root         184  0.0  0.2  17984  8880 ?        Ss   09:26   0:00 /usr/lib/systemd/systemd-logind
root         214  0.0  0.0   3124  1976 tty1     Ss+  09:26   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root         226  0.0  0.5 107044 23192 ?        Ssl  09:26   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-up
root         317  0.0  0.0   3188  1112 ?        Ss   09:26   0:00 /init
root         318  0.0  0.0   3204  1124 ?        S    09:26   0:00 /init
asruthi+     319  0.0  0.1   6080  5364 pts/0    Ss+  09:26   0:00 -bash
root         320  0.0  0.1   6700  4660 pts/1    Ss   09:26   0:00 /bin/login -f
asruthi+     366  0.0  0.2  20332 11564 ?        Ss   09:26   0:00 /usr/lib/systemd/systemd --user
asruthi+     367  0.0  0.0  21176  3572 ?        S    09:26   0:00 (sd-pam)
asruthi+     389  0.0  0.1   6080  5332 pts/1    S+   09:26   0:00 -bash
polkitd     1108  0.0  0.1 308184  7616 ?        Ssl  09:30   0:00 /usr/lib/polkit-1/polkitd --no-debug
syslog      3052  0.0  0.1 222524  5148 ?        Ssl  09:31   0:00 /usr/sbin/rsyslogd -n -iNONE
root        3757  0.0  0.0   3188  1120 ?        Ss   10:05   0:00 /init
root        3758  0.0  0.0   3204  1260 ?        S    10:05   0:00 /init
asruthi+    3761  0.0  0.1   6084  5372 pts/2    Ss   10:05   0:00 -bash
asruthi+    3836  0.0  0.1   8300  4292 pts/2    R+   10:16   0:00 ps aux
asruthi_chowdary@DESKTOP-PQIM4MO:~$ top
top - 10:16:54 up 50 min,  1 user,  load average: 0.00, 0.00, 0
```

```
asruthi_chowdary@DESKTOP-PQIM4MO:~$ top
top - 10:16:54 up 50 min,  1 user,  load average: 0.00, 0.00, 0
Tasks:  25 total,   1 running,  24 sleeping,   0 stopped,   0 z
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,
MiB Mem :   3789.6 total,   3185.2 free,    506.3 used,    240.
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3283.

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
      1 root      20   0   22036  13188   9552 S   0.0   0.3
      2 root      20   0    3180   2204   2072 S   0.0   0.1
      6 root      20   0    3180   2044   1940 S   0.0   0.1
     48 root      19  -1   66840  15984  14756 S   0.0   0.4
    105 root      20   0   25148   6496   5144 S   0.0   0.2
    116 systemd+  20   0   21348  13180  11060 S   0.0   0.3
    117 systemd+  20   0   91040   7944   6976 S   0.0   0.2
    169 root      20   0    4244   2684   2432 S   0.0   0.1
    170 message+  20   0    9780   5664   4732 S   0.0   0.1
    184 root      20   0   17984   8880   7824 S   0.0   0.2
    214 root      20   0    3124   1976   1836 S   0.0   0.1
    226 root      20   0  107044  23192  13732 S   0.0   0.6
    317 root      20   0    3188   1112    980 S   0.0   0.0
    318 root      20   0    3204   1124    980 S   0.0   0.0
    319 asruthi+  20   0    6080   5364   3648 S   0.0   0.1
    320 root      20   0    6700   4660   3876 S   0.0   0.1
    366 asruthi+  20   0   20332  11564   9440 S   0.0   0.3
    367 asruthi+  20   0   21176   3572   1848 S   0.0   0.1
    389 asruthi+  20   0    6080   5332   3668 S   0.0   0.1
   1108 polkitd   20   0  308184   7616   6720 S   0.0   0.2
   3052 syslog    20   0  222524   5148   4236 S   0.0   0.1
   3757 root      20   0    3188   1120    980 S   0.0   0.0
   3758 root      20   0    3204   1260   1108 S   0.0   0.0
   3761 asruthi+  20   0    6084   5372   3652 S   0.0   0.1
   3837 asruthi+  20   0    9296   5672   3504 R   0.0   0.1
```

```
asruthi_chowdary@DESKTOP-PQIM4MO:~$ ps aux | grep -i ssh
asruthi+    3844  0.0  0.0   3972  2100 pts/2    R+   10:17   0:00 grep --color=auto -i ssh
asruthi_chowdary@DESKTOP-PQIM4MO:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       504Mi       3.1Gi       3.5Mi       240Mi       3.2Gi
Swap:          1.0Gi          0B       1.0Gi
asruthi_chowdary@DESKTOP-PQIM4MO:~$
```